Recognition of Emergency Vehicle Sirens and Velocity Estimation

Advanced Driver Assistance Systems Project

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Sound Detection

How can the detection of environmental sounds be important in automotive applications?

- Supporting other ADAS for environmental awareness.
- Detecting specifics sounds (**emergency vehicles**, alarms, voices, ...) to improve safety.
- Increase amount of data for data analysis.

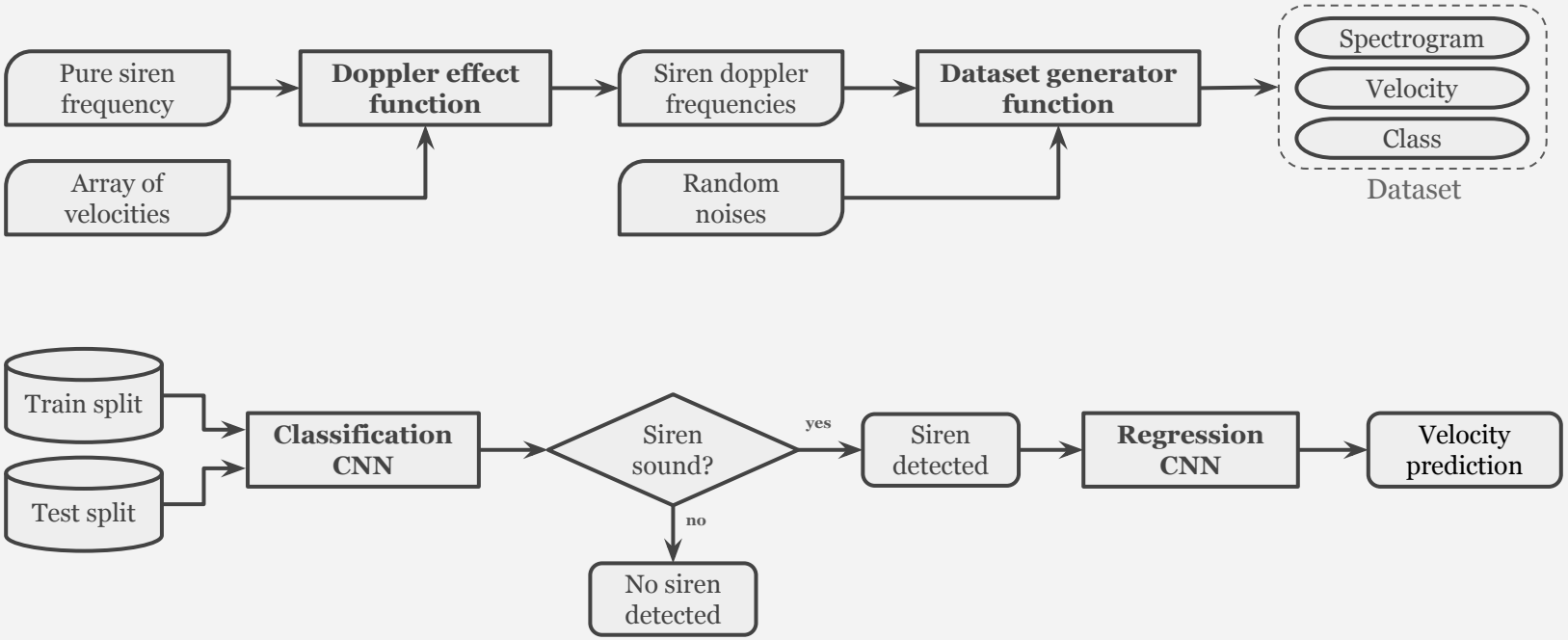
State of the Art

- Microphones: they can only perceive sounds and direction of the source.
- Dataset problem: gathering emergency siren data can be difficult and problematic, so there are virtually no usable (and free) data.
- Deep Learning for sound detection: classification to identify a specific sound and regression to estimate a specific value.
- Spectrograms are the most used images for sound classification in CNNs.

Requirements

- Sensors: microphones to gather the audio data.
- Hardware integration (mounting, calibration, ...) and known location of each microphone on the car for the best efficiency.
- Complete audio data for the training. The dataset must contain realistic data, binary classes to identify sirens and velocity of the emergency vehicle (positive = distancing, negative = approaching).
- Classification for sound recognition.
- Regression for velocity estimation.
- Practical testing in a real life environment.

Architecture

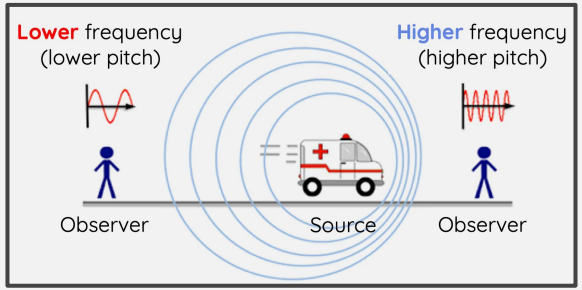


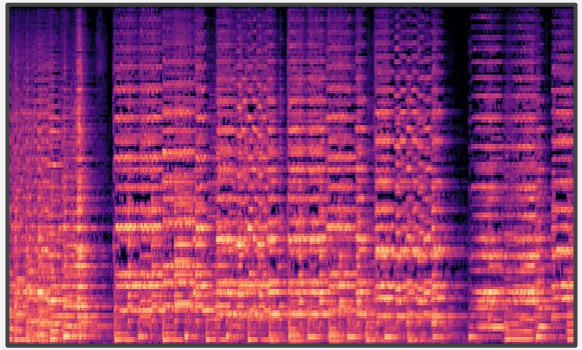
Dataset Generation

- **Doppler effect**: $f_o = \frac{v + v_o}{v + v_s} f_s$

- **Mel-Spectrogram**: Time-frequency visual representation of a sound in a Mel-scale

- **Noise classes**: rain, voices, engines, animals, horns, brakes, constructions, ecc…



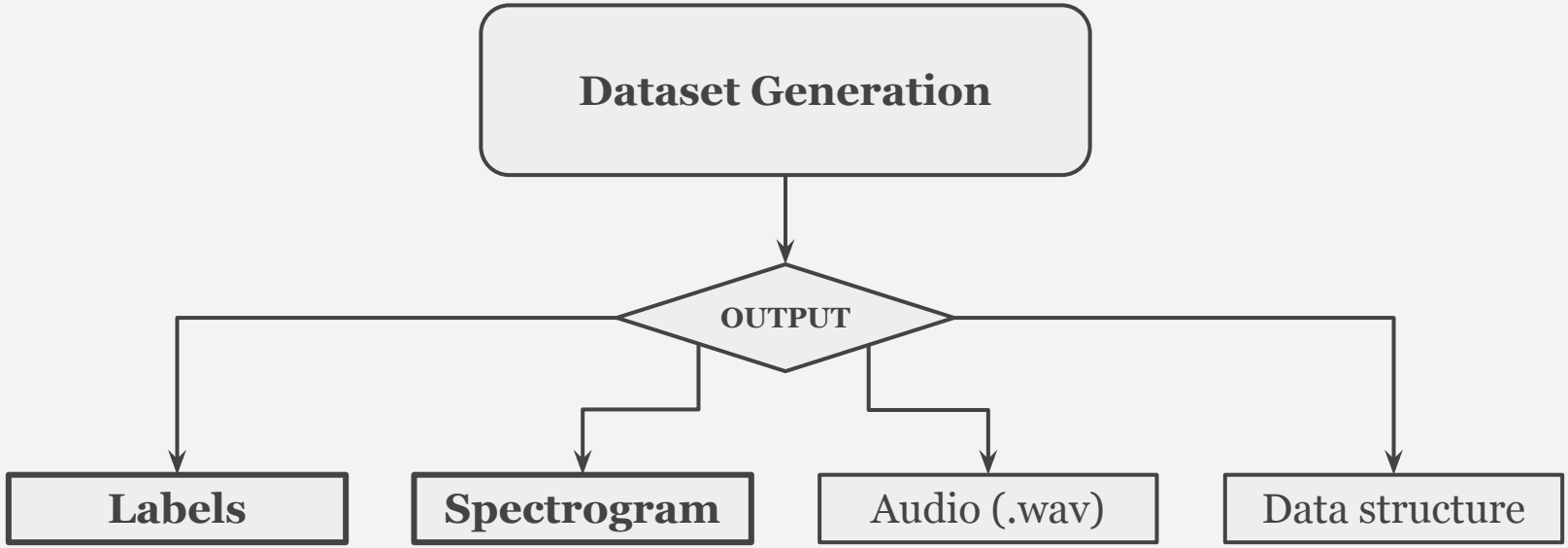


Dataset Generation

Specific data needs → Generation of artificial dataset… How?

- Starting from "pure" siren frequency.
- Simulating movement through Doppler effect
- Adding random noise to the moving siren to simulate a real life urban environment.
- Shifting and cutting the track to improve diversity and standardizing the dimensions.
- Converting the audio files to Mel spectrograms.
- Repeating the process to obtain a large number of combinations.

Output Format

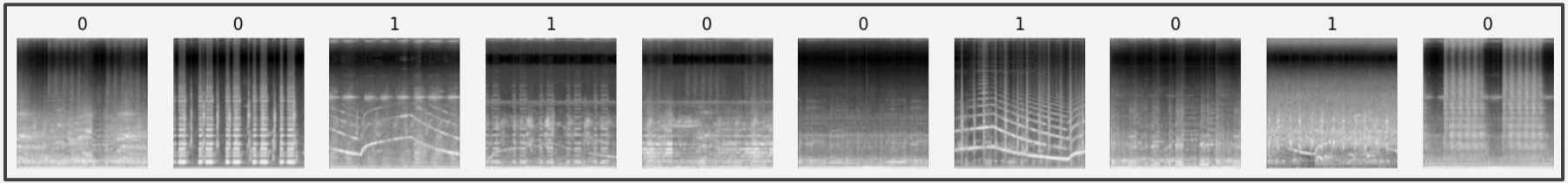


Classification

- Mel spectrogram
- Velocity (positive or negative)

CNN

Predicted class (0,1)



Classification

Hyperparameters

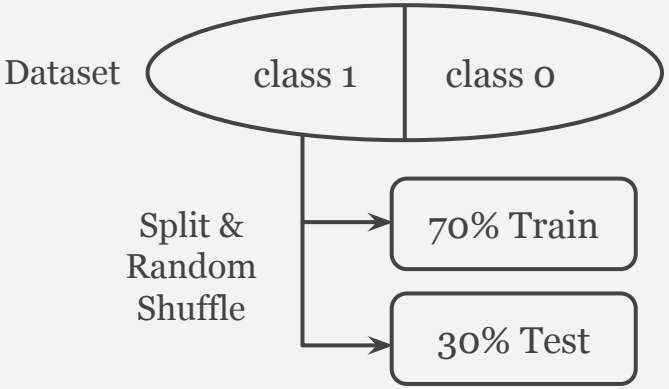
Images : 128x128 (grayscale)

Batch size : 32

Epochs : 20

Learning Rate : 0.001

Loss function : BCE



$$\text{ACCURACY} = \frac{TP + TN}{TP + TN + FP + FN}$$

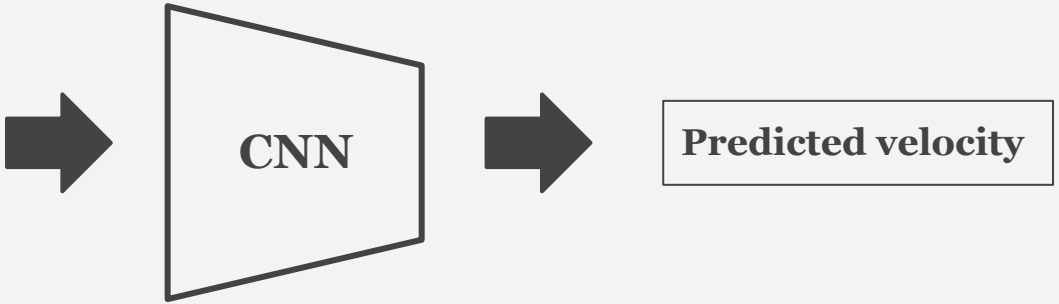
Code

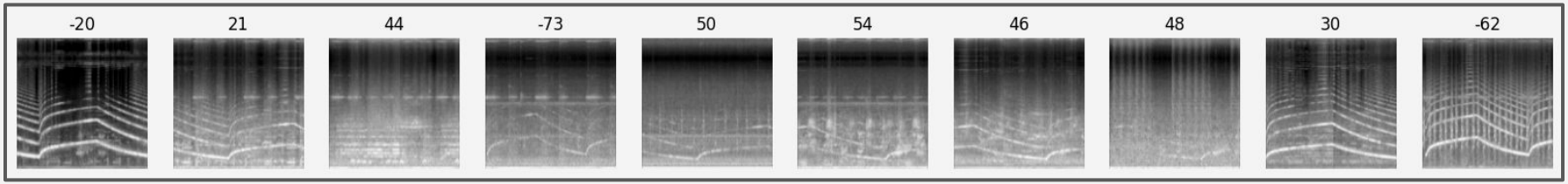
```
self.conv1 = nn.Conv2d(1, 16, 3, 1, 1)
self.conv2 = nn.Conv2d(16, 32, 3, 1, 1)
self.conv3 = nn.Conv2d(32, 64, 3, 1, 1)

self.bn1 = nn.BatchNorm2d(16)
self.bn2 = nn.BatchNorm2d(32)
self.bn3 = nn.BatchNorm2d(64)

self.fc1 = nn.Linear(128 * 128, 512)
self.fc2 = nn.Linear(512, 1)

self.pool = nn.MaxPool2d(2, 2, 0)
self.dropout = nn.Dropout(0.5)
```

```
def forward(self, x):
    x = self.pool(F.relu(self.bn1(self.conv1(x))))
    x = self.pool(F.relu(self.bn2(self.conv2(x))))
    x = self.pool(F.relu(self.bn3(self.conv3(x))))

    x = torch.flatten(x, start_dim=1)

    x = F.relu(self.fc1(x))
    x = self.dropout(x)

    x = torch.sigmoid(self.fc2(x))
    return x
```

Regression

- Mel spectrogram
- Velocity (positive or negative)





Regression

Hyperparameters

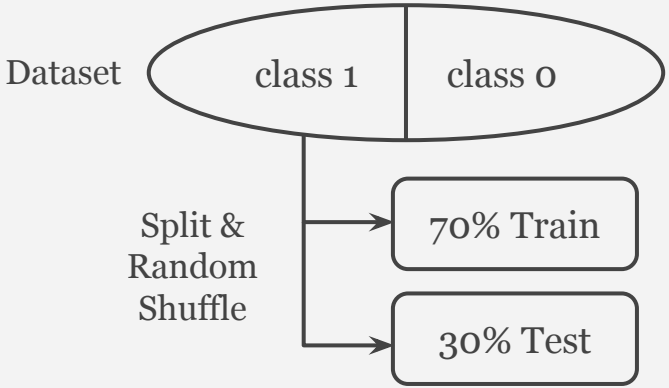
Images : 128x128 (grayscale)

Batch size : 32

Epochs : 20

Learning Rate : 0.001

Loss function : MSE



$$\text{SMAPE} = \frac{1}{n}\sum_{t=1}^{n}\frac{|F_t - A_t|}{(A_t + F_t)/2}$$

Code

```python
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self.conv3 = nn.Conv2d(32, 64, 3, 1, 1)

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self.fc2 = nn.Linear(512, 1)

self.pool = nn.MaxPool2d(2, 2, 0)
self.dropout = nn.Dropout(0.5)
```
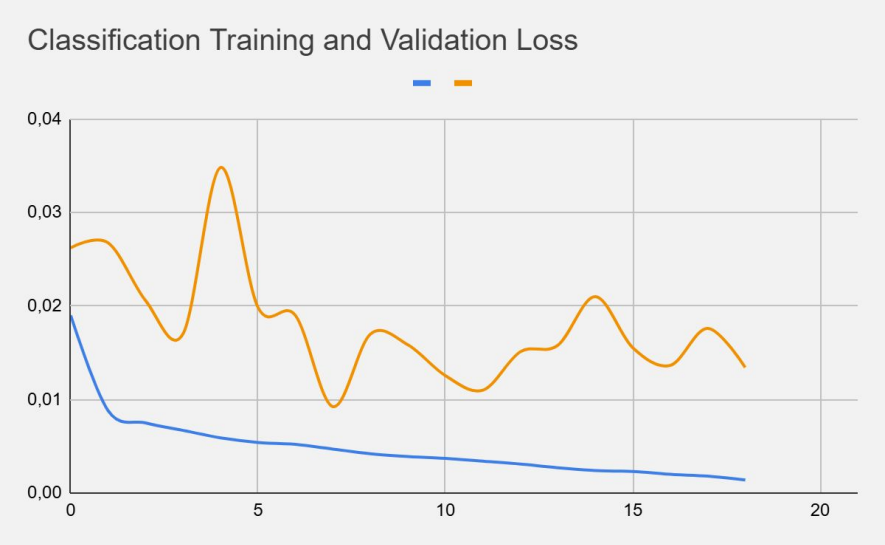
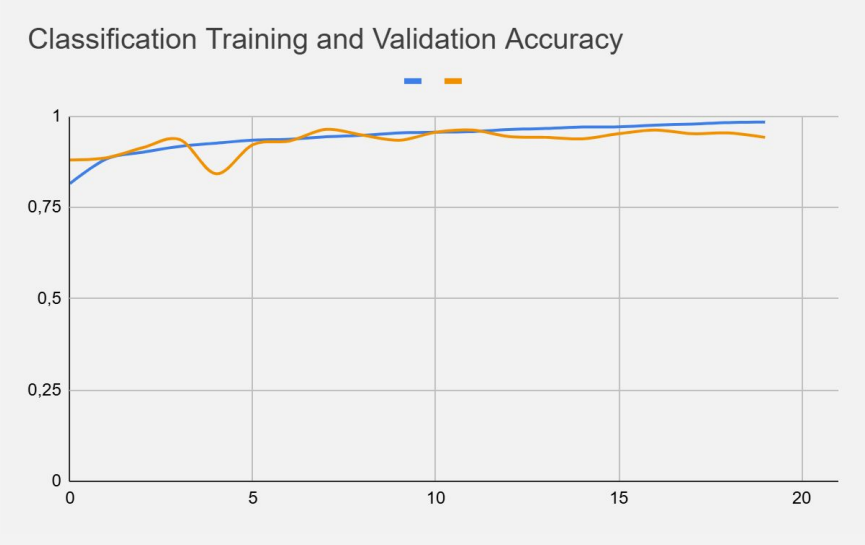

```python
def forward(self, x):
    x = self.pool(F.relu(self.bn1(self.conv1(x))))
    x = self.pool(F.relu(self.bn2(self.conv2(x))))
    x = self.pool(F.relu(self.bn3(self.conv3(x))))

    x = torch.flatten(x, start_dim=1)

    x = F.relu(self.fc1(x))
    x = self.dropout(x)

    x = self.fc2(x) # NO SIGMOID
    return x
```

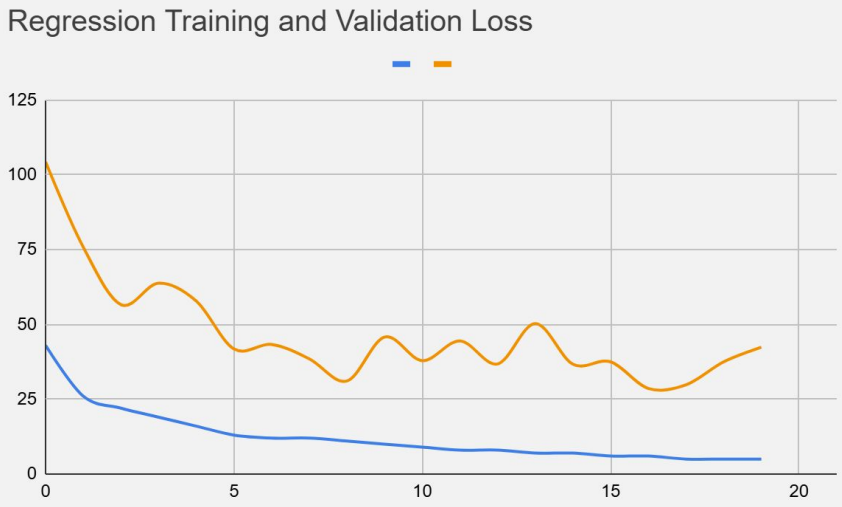
Results

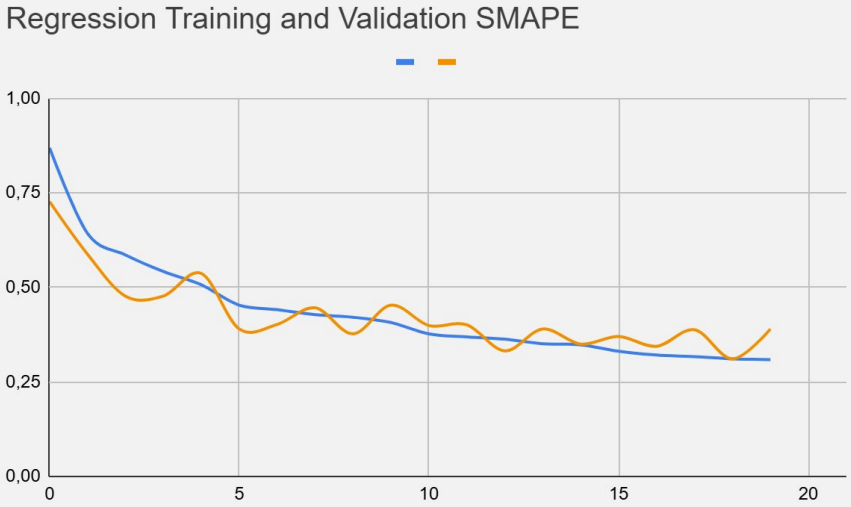




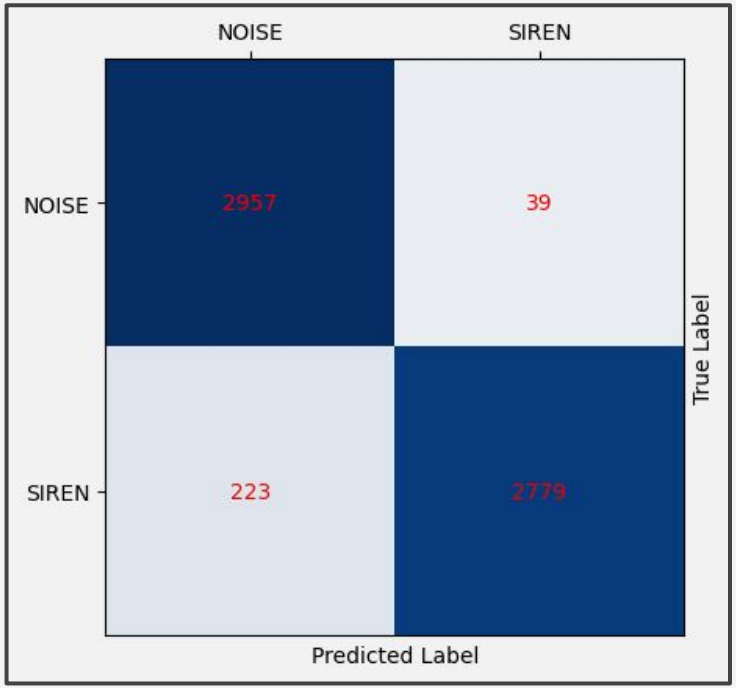
Variance in the validation metrics caused by small sample size (only 500 images/epoch to reduce training times)

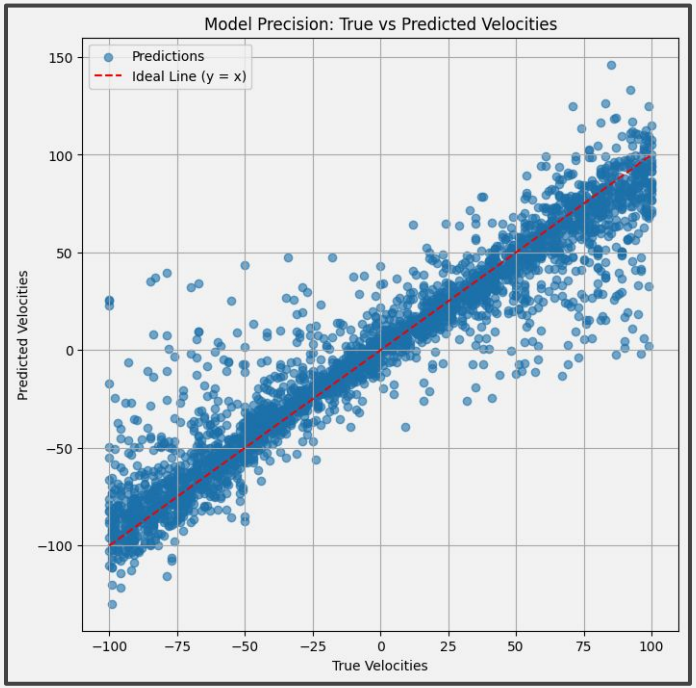
Results


Regression Training and Validation Loss
125
100
75
50
25
0
0
5
10
15
20


Regression Training and Validation SMAPE
1,00
0,75
0,50
0,25
0,00
0
5
10
15
20

Results


NOISE
SIREN
NOISE
SIREN
2957
39
223
2779
True Label
Predicted Label


Model Precision: True vs Predicted Velocities
Predictions
Ideal Line (y = x)
150
100
50
0
−50
−100
Predicted Velocities
−100
−75
−50
−25
0
25
50
75
100
True Velocities

Assumptions and Improvements

- Assumption of moving source and static receiver (can be solved by adding doppler to every sound).
- In a realistic scenario, building, weather and unpredictable events, can obstruct the sound wave and give false measurements.
- Adding more noises and different types of sirens to achieve better results in a real scenario.
- Array of microphones for sound direction estimation: (DeepEar).

Conclusions

Dataset:

- Faster and easier compare to manual acquisition
- Possibility of generating every possible sound combination
- Generate data using multiple formats

Networks:

- The network allowed to identify features using a Mel spectrogram of the desired sound
- Acceptable performances for developing phase, but not enough for real life applications

Thanks for the attention!

References

- DeepEar: Sound Localization With Binaural Microphones,Qiang Yang , Student Member, IEEE and Yuanqing Zheng https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9954178.

- Sound Classification Using Convolutional Neural Networks, Kaustumbh Jaiswal, Dhairya Kalpeshbhai Pate https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8648629

- Acoustic Localization of Emergency Vehicle Sirens in Traffic, Mattes Ohlenbusch1,2, Christian Rollwage1, Joerg Bitzer https://pub.dega-akustik.de/DAGA_2020/data/articles/000010.pdf